

Practice problems #2

1. Mark your answer by True or False. Justify your answer.

(a) The matrix A is row equivalent to B where

$$A = \begin{bmatrix} 0 & -2 & 5 \\ 1 & 4 & -7 \\ 3 & -1 & 6 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 1 & 4 & -7 \\ 0 & -2 & 5 \\ 3 & -1 & 6 \end{bmatrix}.$$

(b) The matrix A is row equivalent to B where

$$A = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 4 & -1 & 3 & -6 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 0 & 7 & -1 & -6 \end{bmatrix}.$$

(c) If a matrix A is row equivalent to B , then the matrix B is row equivalent to A .

(d) The augmented matrices $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$ are equivalent.

Answers or Hints are on the back.

Answers:

#1(a): T. Justify by specifying row operations.

#1(b): T. Justify by specifying row operations.

#1(c): T. Explain why!

#1(d): F. Explain why!